Thompson Sampling with Beta Distributions

A Deep Dive into Reinforcement Learning, Bayesian Updating, and Exploration

Introduction

One of the most important problems in Reinforcement Learning is balancing:

- **Exploration** — trying options we know little about.
- **Exploitation** — choosing options that currently seem best.

A classic example is the **Multi-Armed Bandit** problem.

Imagine a slot machine with multiple levers (arms):

Arm	True Success Rate
A	Unknown
B	Unknown
C	Unknown

Every time we pull an arm:

- Success → Reward = 1
- Failure → Reward = 0

Our goal is to maximize rewards while learning which arm is best.

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1. Why Thompson Sampling?

Many algorithms require manually specifying how much exploration to perform.

Examples include:

- ε-Greedy
- Upper Confidence Bound (UCB)

Thompson Sampling takes a different approach.

Instead of maintaining a single estimate for each arm, it maintains a **probability distribution** representing uncertainty.

This allows exploration to happen naturally.

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2. The Unknown Success Probability

Suppose an arm has a true success probability:

\theta

Examples:

- $\theta = 0.2$ → succeeds 20% of the time
- $\theta = 0.5$ → succeeds 50% of the time
- $\theta = 0.8$ → succeeds 80% of the time

The challenge is that we do not know θ.

Instead, we maintain a belief about possible values of θ.

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3. Why Use the Beta Distribution?

The Beta distribution is ideal because:

- It is defined between 0 and 1.
- It naturally models probabilities.
- It updates easily after observing successes and failures.

We represent our belief as:

$$\theta \sim Beta(\alpha, \beta)$$

This means:

"Our belief about θ follows a Beta distribution with parameters α and β."

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4. Understanding α and β

The parameters represent accumulated evidence.

Interpretation

- α = evidence supporting success
- β = evidence supporting failure

Starting Prior

Typically we start with:

Beta(1,1)

This represents complete uncertainty.

Updating After Observations

After observing:

- S successes
- F failures

The posterior becomes:

Beta(1 + S, 1 + F)

Example

Observed outcomes:

Success

Success

Failure

Failure

Failure

Then:

- Successes = 2
- Failures = 3

Therefore:

- $\alpha = 3$
- $\beta = 4$

Result:

Beta(3,4)

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5. The Beta Distribution Formula

The probability density function (PDF) is:

$$f(x;\alpha,\beta) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$$

where:

$0 \le x \le 1$

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6. What Does x Mean?

This is one of the most important concepts.

The x-axis represents:

the possible true success probability θ.

Examples:

x	Meaning
0.1	Success rate = 10%
0.5	Success rate = 50%
0.9	Success rate = 90%

The x-axis is **not**:

- Time
- Round number
- Reward count

It represents:

"A possible value of the arm's true success probability."

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7. What Does f(x) Mean?

The y-axis represents density.

It answers:

"How plausible is this value of θ?"

Example:

If:

- f(0.7) = 3
- f(0.2) = 0.5

Then:

- 70% success probability appears more plausible.
- 20% success probability appears less plausible.

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8. Understanding the Shape of the Curve

Beta(1,1)

Uniform distribution.

No information.

Every success probability is equally likely.

Beta(10,2)

Many successes.

Distribution shifts toward 1.

Interpretation:

"High success probabilities are more believable."

Beta(2,10)

Many failures.

Distribution shifts toward 0.

Interpretation:

"Low success probabilities are more believable."

9. What Is the Beta Function B(α,β)?

The denominator:

B(α,β)

is called the Beta Function.

Defined as:

$$B(\alpha,\beta) = \int_0^1 t^{\alpha-1}\,(1-t)^{\beta-1}dt$$

Its purpose is normalization:

The total area under the probability density curve equals 1.

For intuition behind Thompson Sampling, this term can generally be ignored.

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10. Mean of the Beta Distribution

The expected success probability is:

$$E[\theta] = \frac{\alpha}{\alpha+\beta}$$

Example

For Beta(3,4):

Mean = 3 / 7 = 0.429

Estimated success rate ≈ 42.9%.

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11. Common Misunderstanding

Many people think Thompson Sampling works as follows:

1. Compute the Beta distribution.
2. Find its highest point.
3. Choose that value.

This is **not** Thompson Sampling.

That would simply be using the mean or mode.

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12. What Thompson Sampling Actually Does

Suppose:

$\theta \sim \text{Beta}(3,4)$

Instead of choosing:

$\theta = 0.429$

we draw a random sample:

$\tilde{\theta} \sim \text{Beta}(3,4)$

Possible samples:

- 0.25
- 0.39
- 0.51
- 0.61

Each draw is different.

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13. Multi-Arm Example

Suppose:

Arm A:

Beta(81,21)

Arm B:

Beta(2,1)

Arm A

Mean ≈ 0.79

Lots of data.

Very confident.

Typical samples:

- 0.78
- 0.79
- 0.80
- 0.81

Arm B

Mean ≈ 0.67

Very little data.

Highly uncertain.

Possible samples:

- 0.12
- 0.31
- 0.76
- 0.95
- 0.99

Large variation.

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14. Why Less-Explored Arms Get Chances

This is the key insight.

When:

$\alpha + \beta$

is small,

there is little data.

Little data implies:

- High uncertainty
- Wide Beta distribution
- Greater variation in samples

Thus unexplored arms occasionally produce very large sampled values.

Example:

Arm	Sampled Value
A	0.80
B	0.95

Arm B wins and gets explored.

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15. Exploration Emerges Automatically

Thompson Sampling never explicitly says:

"Explore 10% of the time."

Instead:

Less Data

↓

More Uncertainty

↓

Wider Beta Distribution

↓

More Extreme Samples

↓

More Exploration

Exploration arises naturally from uncertainty.

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16. Variance of the Beta Distribution

Variance measures uncertainty.

The formula is:

$$Var(\theta) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

Important observation:

As:

$\alpha + \beta$

increases,

variance decreases.

Meaning:

More observations

↓

Less uncertainty

↓

Narrower distribution

↓

More exploitation

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17. Full Thompson Sampling Algorithm

Initialization

For every arm:

- $\alpha = 1$
- $\beta = 1$

Step 1: Sample

For each arm:

Draw:

$\tilde{\theta}_i \sim \mathrm{Beta}(\alpha_i, \beta_i)$

Step 2: Select

Choose the arm with the largest sampled value:

$\text{argmax}(\theta_i)$

Step 3: Observe Reward

- Success → Reward = 1
- Failure → Reward = 0

Step 4: Update

If success:

$\alpha_i = \alpha_i + 1$

If failure:

$\beta_i = \beta_i + 1$

Repeat indefinitely.

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18. Bayesian Interpretation

Each arm has an unknown parameter:

θ_i

The Beta distribution represents our belief about that parameter.

Every round:

1. Build a belief.
2. Sample a possible world.
3. Choose the best action in that world.
4. Observe reality.
5. Update the belief.

Flow:

Belief

↓

Sample Possible World

↓

Choose Best Action

↓

Observe Reality

↓

Update Belief

↓

Repeat

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19. Key Takeaways

What is the x-axis?

The possible true success probability:

$0 \leq \theta \leq 1$

What is the y-axis?

The plausibility (density) of each success probability.

What are α and β?

Evidence counts:

- $\alpha \rightarrow$ success evidence
- $\beta \rightarrow$ failure evidence

Why do unexplored arms get chances?

Because they have greater uncertainty.

Greater uncertainty creates wider distributions.

Wider distributions occasionally generate large samples.

What does Thompson Sampling sample?

It does **not** sample:

- Reward
- Mean
- Mode

It samples:

$\theta \sim \text{Beta}(\alpha, \beta)$

for each arm.

Why is Thompson Sampling powerful?

It automatically balances:

- Exploration
- Exploitation

without requiring a manually chosen exploration rate.

Final Mental Model

Think of each arm as saying:

"I believe my success rate could be anywhere within this range."

The Beta distribution captures that belief.

Thompson Sampling repeatedly asks:

"Given my current beliefs, what if this arm is actually the best?"

and uses random sampling to answer that question.

As more data arrives:

- Uncertainty shrinks.
- Beliefs become sharper.
- Exploration decreases.
- The algorithm naturally converges toward the best arm.

What does

$$\theta \sim \text{Beta}(3,4)$$

mean?

It means:

The unknown probability θ is modeled as a **random variable** that follows a Beta distribution with parameters 3 and 4.

So instead of saying:

- "$\theta = 0.6$" (one fixed unknown value)

we say:

- "θ could be many values between 0 and 1, and we assign probabilities to each possible value."

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2. What is θ?

θ represents:

The true success probability of an action

Example:

- Probability of click
- Probability of winning a game
- Probability of reward = 1

So:

- $\theta = 0.7 \rightarrow$ 70% success rate
- $\theta = 0.2 \rightarrow$ 20% success rate

But we **don't know it exactly**, so we model uncertainty.

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3. What is Beta?

The **Beta distribution** is a probability distribution over probabilities.

That sounds circular, so here is the simple meaning:

Beta describes how likely different values of θ(between 0 and 1) are.

So it is a distribution defined on:

$$0 \leq \theta \leq 1$$

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4. What does Beta(3,4) specifically mean?

$$\text{Beta}(3,4)$$

means:

- We have seen **2 successes + prior**
- We have seen **3 failures + prior**
- So we are slightly more biased toward lower probabilities

Shape intuition

Beta(3,4) looks like:

- Peak around ~0.4
- More weight on lower values than higher values

So it says:

"Success probability is likely around 40%, but we are uncertain."

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5. What is the formula?

The Beta distribution density is:

$$f(x; \alpha, \beta) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$$

For Beta(3,4):

$$f(x) = \frac{x^2(1-x)^3}{B(3,4)}$$

This function tells:

How plausible each value of x(i.e., each possible θ) is.

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6. What does the x-axis mean?

For Beta(3,4):

- x-axis = possible values of θ
- from 0 to 1

Example:

x	Meaning
0.1	“success rate is 10%”
0.4	“success rate is 40%”
0.8	“success rate is 80%”

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7. What does Beta represent in one sentence?

Beta represents our uncertainty about a probability.

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8. Why do we need Beta in RL?

Because in reinforcement learning:

We don’t know true reward probabilities.

So instead of guessing a single number, we maintain:

- many possible values of θ

- and how likely each one is

That is exactly what Beta gives us.

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9. Key intuition (very important)

Compare:

Case 1

$$\mathrm{Beta}(1,1)$$

→ No knowledge
→ All probabilities equally likely

Case 2

$$\mathrm{Beta}(3,4)$$

→ Some experience
→ Slight belief that success rate is moderate (~0.4)

Case 3

$$\mathrm{Beta}(100,10)$$

→ Strong evidence
→ Very confident success rate is high (~0.9)

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10. One-line summary

$$\theta \sim \text{Beta}(3,4)$$

means:

"The true success probability $\theta$is unknown, but based on data, values around 0.4 are more likely than others, and uncertainty is still present."

If you want next step, I can show you:

- how Beta updates after every RL step (step-by-step table), or
- how Thompson Sampling actually picks actions using this distribution visually.

Thompson Sampling with Beta Distribution (Intuition)

1. Updating the Beta Distribution After Each Round

After every round, we update our belief about each arm using a Beta distribution:

- If success →

$$\alpha \leftarrow \alpha + 1$$

- If failure →

$$\beta \leftarrow \beta + 1$$

So each arm is represented as:

$$\theta \sim \text{Beta}(\alpha, \beta)$$

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2. Shape of the Beta Distribution

The Beta distribution reflects uncertainty:

- **More explored arms** → sharper (narrow) curve
- **Less explored arms** → wider curve

This happens because variance decreases as data increases:

$$\text{Var}(\theta) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

Intuition:

- Few observations → high uncertainty → wide curve
- Many observations → low uncertainty → sharp curve

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3. What "Sampling from the Curve" Really Means

We do NOT pick random points inside the curve.

Instead, we do:

$$\theta_i \sim \text{Beta}(\alpha_i, \beta_i)$$

This means:

- We randomly sample a value according to the probability distribution
- High-probability regions are more likely to be selected
- Low-probability regions are less likely to be selected

So each arm produces one sampled value per round.

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4. Action Selection Rule (Important Correction)

We do NOT pick the maximum x-value from the curve.

Instead, we do:

Step-by-step:

For each arm:

- Draw one sample:

$$\theta_i \sim \text{Beta}(\alpha_i, \beta_i)$$

Then choose:

$$\text{arm} = \arg\max_i (\theta_i)$$

Example:

Arm	Sampled Value
A	0.72
B	0.65
C	0.81

→ Choose Arm C (highest sampled value)

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5. Full Thompson Sampling Algorithm

Repeat every round:

Step 1: Maintain beliefs

Each arm has:

$$\text{Beta}(\alpha_i, \beta_i)$$

Step 2: Sampling

For each arm:

$$\theta_i \sim \text{Beta}(\alpha_i, \beta_i)$$

Step 3: Selection

Choose the arm with the highest sampled value:

$$\arg\max_i \theta_i$$

Step 4: Observe reward

- Success → increase α
- Failure → increase β

Step 5: Repeat

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6. Key Intuition Behind Exploration vs Exploitation

Your idea:

Better arms get sharper curves and are exploited more

✔ This is correct, but the deeper mechanism is:

Why exploitation happens:

- Good arms → higher mean
- Good arms → more data → less uncertainty
- Therefore → consistently high samples

Why exploration happens automatically:

Even poorly explored arms:

- Have wide distributions
- Occasionally produce very high samples
- Sometimes beat better arms in sampling

So they still get selected occasionally.

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7. The Most Important Insight

Thompson Sampling does NOT explicitly force exploration.

Instead:

Exploration emerges naturally from uncertainty.

- Uncertain arms → wide Beta → random high samples
- Certain arms → narrow Beta → stable samples

This balance automatically leads to optimal learning behavior.

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8. Final Mental Model

Think of each arm as saying:

"My true success rate could be anywhere in this range, but I'm not fully sure."

Each round:

1. Every arm imagines one possible reality (sampling)
2. You choose the best imagined outcome
3. You observe reality
4. You update beliefs
5. Repeat

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